ROADMAP PROJECT 1

CLOUD COMPUTING INTRO:

Cloud computing is the delivery of computing services over the internet. Computing services include common IT infrastructure such as virtual machines, storage, databases, and networking.

ON PERMISES INFRASTUCTURE INTRO:

On-premises infrastructure refers to IT systems, servers, networking equipment, storage, and software that are physically located and managed within an organization's own building or data center, instead of being hosted in the cloud.

CLOUD Advantages & Disadvantages:

Cloud	
Advantage	Disadvantage
Availability and Scalability. 99% available varies depending on service, look at SLA for details. Scalability 2 types vertical, horizontal. Vertical scaling is focused on increasing or decreasing the capabilities of resources. Horizontal scaling is adding or subtracting the number of resources.	**Internet Dependency** If internet connectivity is unavailable, access to cloud-hosted applications may be disrupted.
Reliability and Predictability Reliability is the ability of a system to recover from failures and continue to function. It's also one of the pillars of the Microsoft Azure Well-Architected Framework.Cost predictability is focused on predicting or forecasting the cost of the cloud spend	**Less Hardware Control** You cannot choose the exact physical server or storage hardware.
Security and Governance.	

Manageability in cloud Automatically scale resource deployment based on need.	
Backup & Disaster Recovery	

What is Azure?

Azure gives you the freedom to build, manage, and deploy applications on a massive global network using your favorite tools and frameworks. Azure's core architectural components can be broken down into two main groupings: the physical infrastructure and the management infrastructure.

Azure subscriptions In Azure, subscriptions are a unit of management, billing, and scale. Subscriptions let you organize resource groups and control billing separately from access.

• Billing boundary: Determines how an Azure account is billed. You can create multiple subscriptions for different billing requirements. Azure generates separate billing reports and invoices for each subscription.

• Access control boundary: Azure applies access-management policies at the subscription level. For example, you might create one subscription for your development work and another for production, each with different spending limits and access rules.

Azure management infrastructure

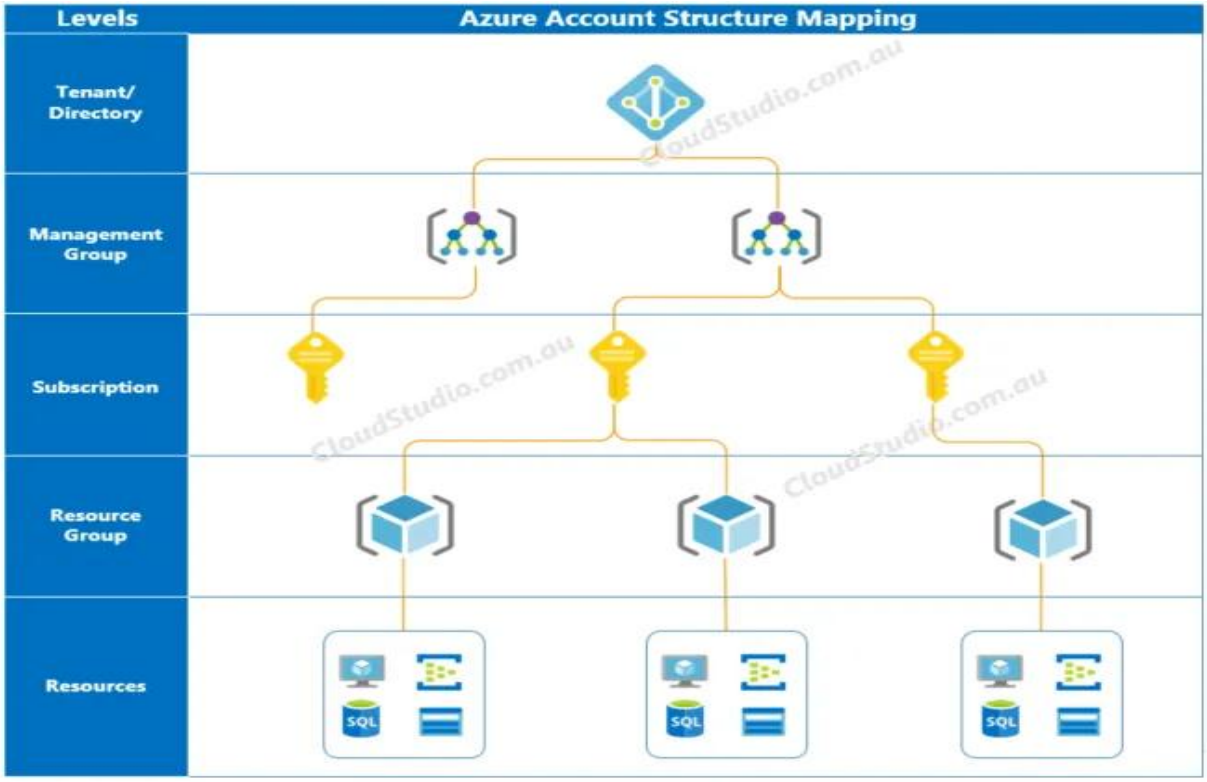
The management infrastructure includes Azure resources and resource groups, subscriptions, and accounts. Understanding this hierarchy helps you organize resources, control who can access what, and manage costs as your Azure usage grows.

Azure resources and resource groups

A resource is the basic building block of Azure. Anything you create, provision, or deploy is a resource. VMs, virtual networks, databases, and Azure AI services are all examples of resources.

Azure management groups

Azure management groups sit above subscriptions. You organize subscriptions into management groups and apply governance conditions — like access policies or compliance rules — to the group. All subscriptions in a management group automatically inherit those conditions, just as resources inherit settings from their resource group.



Azure Blob Storage

Azure Blob Storage is a general-purpose object storage service that stores data as **blobs** inside containers. It is ideal for:

- Best for unstructured data like images, videos, audio, and documents.
- Ideal for application backups, VM disks, and long-term archiving.
- Recommended when you require individual file **soft-delete** restoration capabilities.

Azure Data Lake Storage Gen2 (ADLS Gen2)

ADLS Gen2 is built on top of Blob Storage but adds big data analytics capabilities

through a Hierarchical Namespace (HNS). It is designed for:

- Best for building a data lake or modern data warehouse architecture.
- Ideal when you need to enforce strict, role-based or user-level file and directory permissions.
- Recommended when integrating with heavy data engineering and ELT pipelines (e.g., Azure Data Factory, Snowflake, or Azure Databricks).

Difference Between Blob Storage and ADLS Gen 2

Hierarchical Name Space Structure:

Blob Storage uses a flat structure where folders are simulated in the file path. Azure Blob Storage Gen2 provides a hierarchical namespace that allows you to organize and manage your data more efficiently. This can be particularly useful for big data workloads, where you may have large amounts of data that need to be organized and analyzed in a specific way

Analytics:

ADLS Gen2 is fully optimized for big data analytics engines like Apache Spark, Azure Synapse Analytics, and Databricks. Blob storage lacks these Hadoop-compatible file system optimizations.

Cost:

Azure Blob Storage is a simple and cost-effective storage solution for unstructured data. At the same time, ADLS Gen2 is a more advanced storage solution for big data analytics workloads. The choice between the two depends on your specific requirements and use case.

Few other Differences are **SFTP, Virtual Folder and Serverless SQL.**

Azure Redundancy

Azure redundancy means automatically copying your data to multiple places. This keeps your data safe if there is a hardware crash, power cut, or natural disaster. Redundancy ensures that your storage account meets its availability and durability targets even in the face of failures.

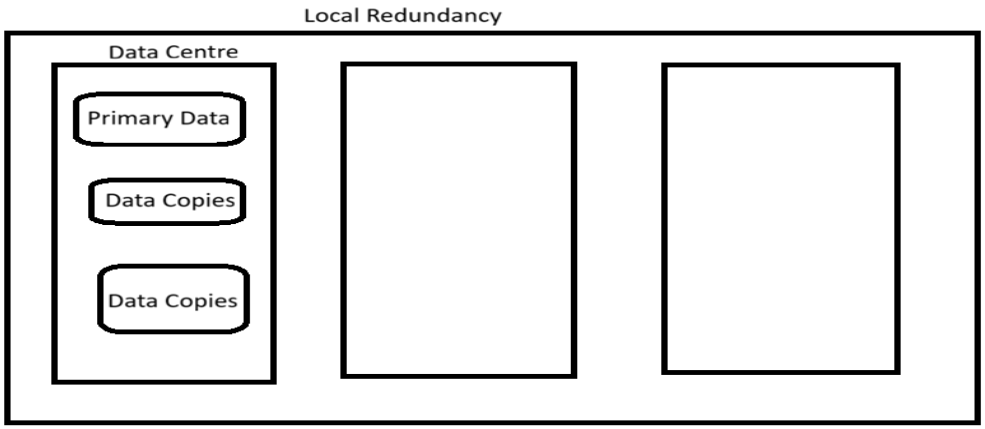
Types of Redundancy

- Local Redundant Storage (LRS)
- Zone-Redundant Storage (ZRS)
- Geo-Redundant Storage (GRS)
- Geo-Zone-Redundant Storage (GZRS)

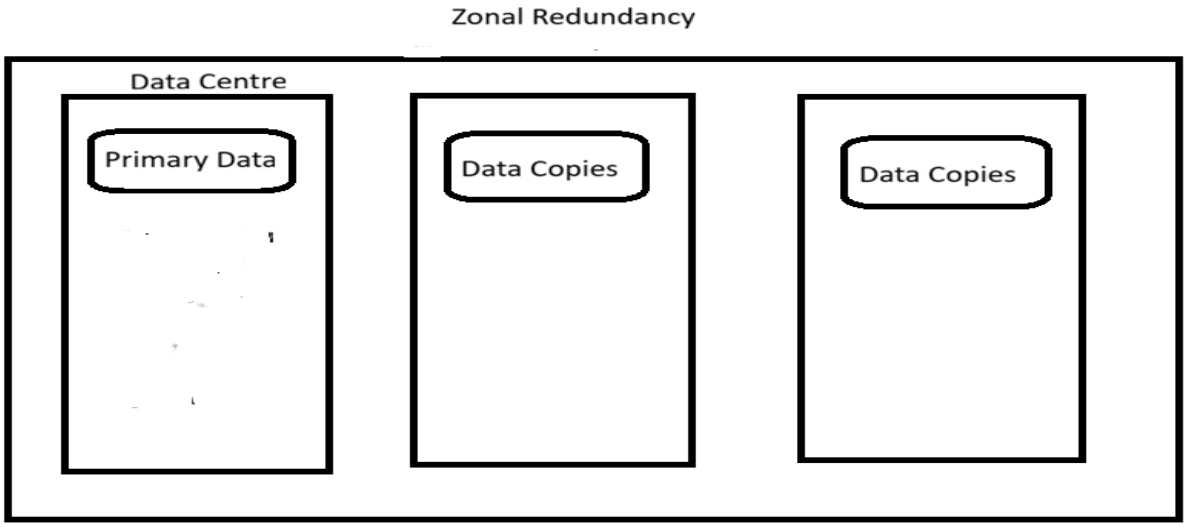
The factors that help determine which redundancy option you should choose include:

- How your data is replicated within the primary region.
- Whether your data is replicated from a primary region to a second, geographically distant region, to protect against regional disasters.
- Whether your application requires read access to the replicated data in the secondary region during an outage in the primary region (geo-replication with read access).

Local redundant storage (LRS) replicates the data within your storage accounts to a single physical datacenter located in the primary region of your choice .



Zone-redundant storage (ZRS) copies your data synchronously across three or more Azure availability zones in the primary region.



Geo-redundant storage (GRS) copies data asynchronously across different geo locations from the primary location. Has 6 copies of replicas.

Geo-zone-redundant storage (GZRS) copies your data synchronously across three or more Azure availability zones in the primary region using ZRS. It then copies your data asynchronously to the secondary region.